CHAPTER TWO

SYSTEM REQUIREMENT SPECIFICATION

2.1 Background Overview

As universities expand in physical size, academic programs, and student population, effective campus navigation becomes an essential support service. Jigjiga University has experienced significant growth in recent years, resulting in an increased number of buildings, colleges, offices, and service facilities distributed across a wide campus area. This expansion, while positive, has introduced challenges related to wayfinding and accessibility for new students, visitors, and even staff members.

Currently, Jigjiga University lacks a centralized digital system that provides structured, accurate, and campus-specific navigation information. New students and visitors often depend on verbal directions, physical signboards, or generic mapping platforms that do not fully represent internal campus routes, building details, or departmental locations. These approaches are inefficient, time-consuming, and prone to errors, leading to confusion, missed classes, late appointments, and overall inconvenience.

With the rapid advancement of web and mobile technologies, universities worldwide are adopting digital campus navigation systems to enhance user experience and improve service delivery. A system specifically designed for the university environment can offer interactive maps, searchable locations, and optimized walking routes tailored to the campus layout. Such a solution not only improves navigation efficiency but also supports institutional modernization and digital transformation initiatives.

In this context, the proposed **UniMap – Smart University Map and Navigation System** aims to address the navigation challenges at Jigjiga University by providing a reliable, user-friendly, and scalable digital platform. The system will serve as a foundational tool to improve accessibility, reduce confusion, and support students, staff, and visitors in navigating the university campus effectively.

2.2 Functional Requirements

Functional requirements describe the specific services and operations that the UniMap system must perform in order to meet user needs. These requirements define how users interact with the system and what functions the system is expected to provide.

2.2.1 User Management

- The system shall allow users to access the application without mandatory registration.
- The system shall allow administrative users to log in securely.
- The system shall provide role-based access for administrators responsible for managing campus map data.

2.2.2 Campus Map Display

- The system shall display an interactive digital map of Jigjiga University campus.
- The system shall show campus buildings, blocks, offices, colleges, and departments on the map.
- The system shall allow users to zoom, pan, and explore different areas of the campus.
- The system shall display building names and brief descriptions when selected.

2.2.3 Search Functionality

- The system shall allow users to search for locations such as buildings, offices, departments, and facilities.
- The system shall provide search results based on keywords entered by the user.
- The system shall highlight the selected location on the campus map.

2.2.4 Route and Navigation

- The system shall provide walking route suggestions between two selected locations within the campus.
- The system shall calculate and display the shortest available route.
- The system shall display step-by-step navigation directions.
- The system shall update routes based on the user's selected starting point and destination.

2.2.5 Facility and Information Services

- The system shall provide additional information about campus facilities such as libraries, cafeterias, dormitories, laboratories, and administrative offices.
- The system shall display images and descriptions of selected buildings or facilities.
- The system shall show important service locations such as registration offices and lecture halls.

2.2.6 Event and Announcement Locations (Optional)

- The system shall display event locations occurring within the university campus.
- The system shall allow administrators to add, update, or remove event location data.

2.2.7 Administrative Management

- The system shall allow administrators to add new buildings, routes, and locations.
- The system shall allow administrators to update existing map data.
- The system shall allow administrators to remove outdated or incorrect information.
- The system shall store all map-related data in a centralized database.

2.3 Non-Functional Requirements

Non-functional requirements define the quality attributes, constraints, and operational conditions of the UniMap system. These requirements ensure that the system performs efficiently, securely, and reliably while providing a good user experience.

2.3.1 Performance Requirements

- The system shall load the campus map within an acceptable time under normal network conditions.
- The system shall respond to user search requests without noticeable delay.
- The routing functionality shall generate walking routes within a short response time.
- The system shall support multiple users accessing the system simultaneously.

2.3.2 Usability Requirements

- The system shall provide a simple, intuitive, and user-friendly interface.
- The system shall be easy to use for first-time users without requiring prior training.
- The system shall use clear labels, icons, and visual indicators for map navigation.
- The system shall support both desktop and mobile devices.

2.3.3 Reliability Requirements

- The system shall be available during normal university working hours.
- The system shall handle user errors gracefully without system failure.
- The system shall ensure data consistency and accuracy of campus location information.

2.3.4 Security Requirements

- The system shall restrict administrative functions to authorized users only.
- The system shall protect stored data from unauthorized access.
- The system shall ensure secure authentication for administrative users.
- The system shall prevent unauthorized modification of map and location data.

2.3.5 Maintainability Requirements

- The system shall be designed in a modular manner to allow easy maintenance.
- The system shall allow administrators to update campus data without modifying system code.
- The system shall support future enhancements such as additional campuses or features.

2.3.6 Scalability Requirements

- The system shall be scalable to accommodate an increasing number of users.
- The system shall allow the addition of new buildings, facilities, and routes.
- The system shall support expansion to other universities if required.

2.3.7 Compatibility Requirements

- The system shall operate on modern web browsers.
- The system shall be compatible with Android devices through mobile deployment.
- The system shall function properly across different screen sizes and resolutions.

2.4 Feasibility Study

The feasibility study evaluates the practicality of developing and implementing the UniMap, Smart University Map and Navigation System. This analysis determines whether the proposed system is achievable within the given time, resources, and technical constraints.

2.4.1 Technical Feasibility

The UniMap system is technically feasible as it will be developed using well-established and widely supported web and mobile technologies. The required development tools, frameworks, and platforms are readily available and supported by extensive documentation and community resources.

The system will be implemented using a modern web-based architecture, allowing it to run efficiently on standard browsers and be packaged for mobile deployment. Mapping technologies and open-source map data can be integrated to represent the university campus accurately. The developer possesses sufficient background knowledge in backend development, APIs, and web technologies, making the implementation technically achievable within the project timeline.

Furthermore, the system does not require advanced hardware or specialized infrastructure. It can be hosted on standard cloud or local servers, making deployment and maintenance manageable.

2.4.2 Economic Feasibility

The proposed system is economically feasible because it relies primarily on open-source software and free development tools. No licensing costs are required for the core technologies, development environments, or databases.

Development will be carried out by a single student as part of an academic project, eliminating labor costs. Hosting and deployment costs can be minimized by using free or low-cost cloud platforms during development and testing phases. Therefore, the overall financial cost of the project is minimal and within acceptable limits for an academic environment.

2.4.3 Operational Feasibility

The UniMap system is operationally feasible as it addresses a real and existing problem within Jigjiga University. The system is designed to be simple and user-friendly, ensuring easy adoption by students, staff, and visitors without requiring training.

Administrative users will be able to manage and update campus data through a secure interface, ensuring that the system remains accurate and up to date. The proposed system integrates smoothly into the existing university environment without disrupting current operations.

Overall, the UniMap system is practical, useful, and acceptable to end users, making it operationally feasible for deployment within Jigjiga University.